



End Term (Even) Semester Regular/Back Examination May 2026

Roll no... 2492001

Name of the Program and semester: BCA, 6th semester

Name of the Course: Mobile Communication

Course Code: TBC-604(2)

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a. Explain multiple access protocols (FDMA, TDMA, CDMA) with diagrams and comparison. (CO2)
- b. Discuss handoff mechanisms and strategies in cellular networks with examples. (CO3)
- c. In a cellular communication system with hexagonal cell geometry, the cell radius is $R=1.5$ km. The system initially uses a cluster size of $N=7$. Due to interference issues, the cluster size increased to $N=12$. Assume the path loss exponent $n=4$. Find the frequency reuse ratio Q , (b) the distance between co-channels cells D (c) The ratio of co-channel interference before and after increasing the cluster size. (CO2)

Q2.

(2X10=20 Marks)

- a. A transmitter produces 50 watts of power. Express the transmit power in dBm and dBW. Also, calculate the received power in dBm at a distance of 100 m for a 900 MHz signal using unity gain antennas. Find received power at 10 km. (given: $\log 2 \approx 0.3$, $\log 3 \approx 0.48$, $\log 5 \approx 0.7$; $\log 9 \approx 0.954$) (CO4)
- b. An 880 MHz carrier signal is frequency modulated using a 100 kHz sinusoidal modulating waveform. The peak deviation of the FM signal is 500 kHz. If this FM signal is received by a superheterodyne receiver having an IF frequency of 5 MHz, determine the IF bandwidth necessary to pass the signal. (CO4)
- c. A beam of unpolarized light is incident from air onto a glass surface. Define the Brewster angle and derive the condition for it. Hence, calculate the Brewster angle for glass of refractive index 1.5. Also, explain why the reflected light is completely plane polarized at this angle. (CO4)

Q3.

(2X10=20 Marks)

- a. An antenna operates at 800 MHz and 1600 MHz with dimensions of 1m. Calculate far-field distance for both frequencies, compare the results, and explain the effect of frequency on far-field distance. (CO3)
- b. Discuss future trends and challenges of IoT-enabled wireless communication systems. (CO4)
- c. Explain key concepts of data transmission: Bandwidth, Latency, Throughput, Security, and Congestion. (CO4)

Q4.

(2X10=20 Marks)

- a. Explain IEEE 802.11 standards in detail and discuss their role in modern wireless networks. (CO5)
- b. A transmitter and a receiver are separated by a distance of 10 km. The system operates at a carrier frequency of 900 MHz under free space propagation conditions. The transmitter and



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receiver antenna gains are $G_t = 1$ and $G_r = 2$, respectively. The power received at the receiver is 3.16×10^{-10} W. Determine the effective isotropic radiated power (EIRP) of the transmitter. (CO3)

Q. A transmitter radiates a power of 100 W. The transmitter and receiver antenna gains are $G_t = 2$ and $G_r = 3$. The distance between them is 5 km. The operating frequency is 600 MHz. Assume free space propagation and system loss factor $L = 1$. Find (a) Wavelength (λ) (b) Received power (P_r) (CO5)

Q5.

(2X10=20 Marks)

a. Explain wireless communication systems with a block diagram and discuss their advantages and limitations. (CO1)

b. A cellular system uses a frequency reuse cluster size of $N = 7$ with omnidirectional antennas. To improve capacity, the system is upgraded to 120° sectoring. (CO2)

Find:

(i) Number of sectors per cell

(ii) New co-channel interference reduction factor

(iii) If total available channels are 840, calculate channels per sector before and after sectoring

c. Describe the architecture of WLAN and its working. (CO4)